ALGEBRA 1 · PERSONALIZED PRACTICE

Math Practice

2026-11-01 · 20 questions

I CAN USE INPUT-OUTPUT RULES TO REPRESENT RELATIONSHIPS.

1. I can use input-output rules to represent relationships.

Complete the table.

x	-2	0	4
$y = 3x - 1$	?	?	?

2. I can use input-output rules to represent relationships.

Write a rule.

Inputs 0,1,2 give outputs 4,7,10.

3. I can use input-output rules to represent relationships.

A machine adds 4 and then doubles the result. The output is 18. Find the input.

4. I can use input-output rules to represent relationships.

A function machine triples the input and subtracts 2. What output comes from input 5?

I CAN DISTINGUISH FUNCTIONS FROM RELATIONSHIPS THAT ARE NOT FUNCTIONS.

5. I can distinguish functions from relationships that are not functions.

Does $\{(1, 2), (1, 3), (2, 4)\}$ represent a function? Explain.

6. I can distinguish functions from relationships that are not functions.

Identify the repeated input.

x	-1	2	-1
y	3	5	8

7. I can distinguish functions from relationships that are not functions.

Pairs $(1, 5), (2, 5), (3, 5)$ all have output 5. Is this a function? Explain.

8. I can distinguish functions from relationships that are not functions.

For $y = 2x - 5$, find the output when $x = -1$.

I CAN IDENTIFY THE DOMAIN AND RANGE OF A RELATION.

9. I can identify the domain and range of a relation.

For $f(x) = x + 2$ with inputs $\{-1, 0, 3\}$, list its outputs. Then extend 4, 9, 14, … one term.

10. I can identify the domain and range of a relation.

List the domain.

$(2, 5), (4, 7), (6, 5)$

11. I can identify the domain and range of a relation.

The function $f(x) = \sqrt{x}$ has real outputs. Explain why negative inputs are excluded.

12. I can identify the domain and range of a relation.

List the range.

$(2, 5), (4, 7), (6, 5)$

I CAN CONNECT LINEAR GROWTH IN TABLES, GRAPHS, AND EQUATIONS.

13. I can connect linear growth in tables, graphs, and equations.

x	0	1	2	3
A	1	4	7	10
B	1	4	9	16

Which is linear? Give numerical evidence.

14. I can connect linear growth in tables, graphs, and equations.

The outputs 2, 5, 8, 11 increase by a constant amount. What is the change per step?

15. I can connect linear growth in tables, graphs, and equations.

Write an equation.

Slope 4; y-intercept −2.

16. I can connect linear growth in tables, graphs, and equations.

Find the slope.

Points (1,3) and (5,11)

I CAN CALCULATE AND INTERPRET SLOPE AS A CONSTANT RATE OF CHANGE.

17. I can calculate and interpret slope as a constant rate of change.

One line has slope −2 and another has slope 3. Compare their directions and rates.

18. I can calculate and interpret slope as a constant rate of change.

Find the run.

$(-1,8)$ to $(6,10)$

19. I can calculate and interpret slope as a constant rate of change.

A tank has 12 L at minute 1 and 27 L at minute 6. Find and interpret its average rate of change.

20. I can calculate and interpret slope as a constant rate of change.

Find the slope through $(0,3)$ and $(4,11)$. Then evaluate $f(2)$ for $f(x) = 3x - 1$.